Life Rays & Fragment Chains – A Theory of Encounter

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For someone who once crossed my path,
whose presence became the origin of this theory.

And for the colleagues, friends, and companions
who helped transform intuition into structure
through challenge, questioning,
and refinement.

Contents

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1 Abstract

This preprint develops the mathematical foundation of the Life Ray Theory, expanding the conceptual framework introduced in Part I into a formal system. The goal is to provide the operator structures, notational conventions, and dynamical principles that allow encounter, fragmentation, resonance, and coupling to be represented in a precise and model-ready form. These formulations prepare the groundwork for later computational, systemic, and empirical applications in Parts III and IV.

Preprint – Work in progress – Part I of a four-part research project.

2 Introduction

Part I of this project presented the phenomenological and conceptual architecture of life rays, fragment chains, and shared spaces. Part II marks a transition: it is the first attempt to translate these intuitions into formal mathematical structures. The aim is not to reduce the richness of encounter, but to render its mechanisms expressible through operators, functions, and topological relations.

What follows is therefore a foundational step. It provides the notation, the structural assumptions, and the core transformations upon which later models of resonance, coupling, and dynamical evolution will rely.

In this sense, the present text serves as the bridge between the intuitive origins of the theory and its scientific articulation.

2.1 Motivation and Scope

Part I introduced the core structures of the Life Ray Theory on a descriptive and phenomenological level. Concepts such as life rays, fragments, shared space, and resonant proximity were defined qualitatively and illustrated through structural metaphors. Their function was conceptual clarity, not mathematical rigor.

The purpose of Part II is fundamentally different. Its scope is to provide:

1. a precise notational system for events, fragments, and spaces,
2. formal operators for transduction, integration, and resonance,
3. metrics for proximity, salience, and coupling,
4. a formal salience function expressing weighted event imprinting,

5. the topological assumptions that underlie the geometry of encounter,
6. and a first formalization of resonant asymmetry (humor) as a special case of temporal divergence.

The formulations introduced here are intentionally minimal: they aim to capture the essential structure of interaction without binding the theory to a specific computational model. Later parts will extend these foundations toward simulation, multi-agent architectures, and empirical applications.

In short, the goal of this part is not to complete the theory, but to make it formally expressible.

3 Notation and Basic Objects

Part I introduced the intuitive architecture of life rays, fragments, and shared spaces. In Part II, these notions receive a formal representation. The following definitions establish the basic mathematical objects on which all subsequent operators and metrics will rely.

3.1 Core Formal Structure

We introduce the minimal formal structure underlying the Life Ray Theory. All later operators, metrics, and interpretations are extensions of this core.

3.2 Agents and Own Space

Let K be a set of agents. Each agent $k \in K$ is associated with an own space S_k, representing its internal experiential state space.

3.3 Life Rays

A life ray is defined as a time-parametrized curve

$$\gamma_k : \mathbb{R}_{\geq 0} \to S_k,$$

mapping personal time to internal state.

3.4 Shared Events

An encounter between two agents $i, j \in K$ at time t is modeled as an event

$$E_{ij}(t).$$

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The shared space is defined as the set of all such events:

$$C = \{E_{ij}(t) \mid i, j \in K,\ t \in \mathbb{R}_{\geq 0}\}.$$

3.5 Fragments

For each event and each involved agent, a fragment is generated by a transduction operator

$$T_k : E_{ij}(t) \mapsto F_k.$$

A fragment is minimally represented as

$$F_k = (t, v_k),$$

where t is the event time and v_k an internal state imprint.

3.6 Fragment Chains

The fragment chain of agent k at time t is the ordered sequence

$$\mathcal{F}_k(t) = (F_k^{(1)}, F_k^{(2)}, \dots, F_k^{(n)}),$$

containing all fragments generated up to time t.

This core defines the minimal ontology of the Life Ray Theory: agents, internal state spaces, time-parametrized life rays, shared events, and fragment-based memory structures. All further definitions refine this structure by adding semantic, affective, and dynamical detail.

4 Extended Structures

4.1 Extended Event Structure

While the core model represents an event as a minimal intersection $E_{ij}(t)$, we now enrich events with descriptive and phenomenological parameters:

$$E_{ij} = (t, C, D, I, V, R),$$

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4.2 Geometric Interpretation of Life Rays

The life ray

$$\gamma_k : \mathbb{R}_{\geq 0} \to \mathcal{S}_k$$

may be interpreted both geometrically and temporally.

Geometrically, γ_k represents a trajectory in the agent's experiential state space $\mathcal{S}_k$. Each point $\gamma_k(\tau)$ corresponds to the internal state of agent k at personal time τ. Intersections between γ_i and γ_j in shared space correspond to encounter events E_{ij}.

Temporally, γ_k is a parametrized curve whose parameter $\tau \in \mathbb{R}_{\geq 0}$ denotes the subjective time of the agent. In this sense, a life ray encodes the continuous evolution of experience, while events represent discrete synchronization points between different rays.

Two rays γ_i and γ_j may therefore intersect multiple times in shared space, giving rise to a sequence of encounter events

$$E_{ij}^{(1)}, E_{ij}^{(2)}, E_{ij}^{(3)}, \ldots,$$

as illustrated in Figure 1.

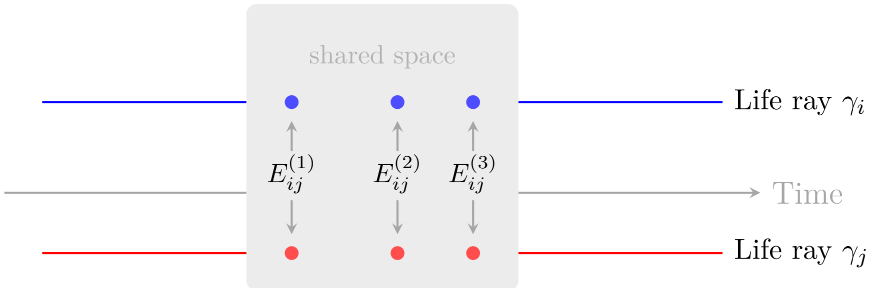

Figure 1: Two life rays (γ_i, γ_j) with repeated intersection events occurring within the shared space.

4.3 Fragments

Every event E_{ij} generates a perspective-dependent memory fragment for each participant:

$$F_k = (s_k, \phi_k, \theta_k, M_k),$$

where s_k is the salience, ϕ_k a semantic label, θ_k the internal timestamp, and M_k a set of affective, bodily, or contextual markers.

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4.4 Fragment Chains

The memory structure of an agent k is modeled as a time-dependent collection of discrete memory fragments. Rather than assuming memory to be continuous, the Life Ray Theory treats biographical memory as an ordered accumulation of such fragments. At any time t, the fragment chain of agent k is represented as an ordered sequence

$$\mathcal{F}_k(t) = (F_k^{(1)}, F_k^{(2)}, \dots, F_k^{(n)}),$$

where each fragment corresponds to a previously transduced encounter event. Together, these fragments define the evolving structure of memory over time.

We denote by $\|$ the concatenation operator on fragment chains, such that

$$\mathcal{F}_k(t+\varepsilon) = \mathcal{F}_k(t) \parallel F_k \tag{1}$$

where $\|$ appends the new fragment F_k to the end of the existing ordered sequence $\mathcal{F}_k(t)$.

Memory formation is assumed to be incremental rather than continuous. When a new salient event occurs, it generates a fragment F_k that is added to the existing chain according to Eq. (1). Here, ε denotes a minimal temporal update step associated with event processing, not a physical duration.

This formulation captures the idea that biographical memory grows through the accumulation of discrete experiential units, while preserving the temporal structure necessary for later integration, resonance, and narrative coherence.

4.5 Shared and Own Space

We distinguish:

- the **shared space** C, in which events E_{ij} occur,
- and the **own space** $\mathcal{S}_k$, containing the internal representation and fragment chain of agent k.

An event E_{ij} lies in shared space, while its transformed fragments F_i and F_j reside in the respective own spaces.

4.6 Time Notation

We use:

$$t \quad \text{(external event time)}, \qquad \theta_k \quad \text{(internal time of fragment formation)},$$

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and, in the case of resonant asymmetry (see Section 7), the distinction between expected and realized meaning times:

$$t_e \neq t_r.$$

These definitions provide the foundational vocabulary from which the formal operators (transduction, integration, resonance), metrics (salience, proximity, coupling), and geometric assumptions of later sections can be expressed.

5 Mechanism: Transduction

5.1 Definition

An *intersection event* $E_{ij}(t, C, D, I, V, R)$ arises in the *shared space* when the life rays of agents i and j interact at time t within a context C (location, task, social setting), with duration D, intensity I, valence V, and perceived reciprocity R. *Transduction* is defined as the operator

$$\mathcal{T}: \; E_{ij}(t, C, D, I, V, R) \; \mapsto \; \big(F_i, \; F_j\big),$$

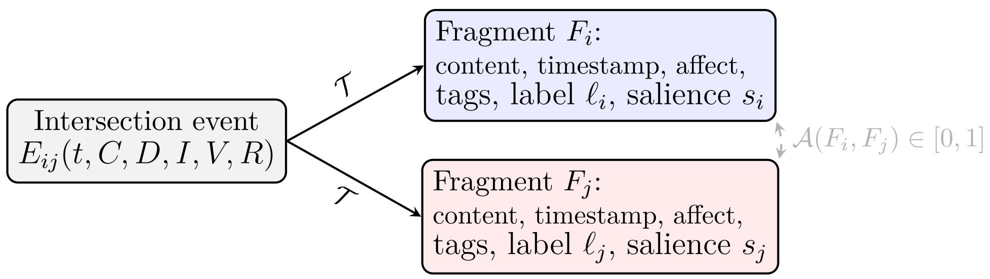

Figure 2: Transduction: The shared event is transformed into two perspective-dependent fragments; an optional alignment $\mathcal{A}$ may occur through shared recall or co-narration.

which transforms the shared event into individual *fragments* F_k (for $k \in \{i, j\}$). A fragment minimally contains the fields

$$F_k = \big(\text{content, timestamp, affect, semantic tags}, \; \ell_k, \; s_k\big),$$

where ℓ_k denotes a semantic label (short interpretation) and $s_k \in \mathbb{R}_{\geq 0}$ represents the fragment's *salience* (imprint strength).

5.2 Process Structure

Transduction unfolds in three conceptual stages:

1. *Detection (Marking):* The event E_{ij} exceeds an activation threshold θ and is registered as salient.

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2. *Encoding (Bundling):* Sensory input and evaluation are condensed into a meaning-bearing core, forming F_k (including tags and semantic label ℓ_k).
3. *Integration (Chain Insertion):* The fragment F_k is appended to the fragment chain of agent k; consolidation processes may increase or decrease s_k over time.

5.3 Asymmetry and Alignment

Transduction is inherently perspective-dependent: from the same event E_{ij}, two generally distinct fragments F_i and F_j emerge, differing in semantic label ℓ and salience s. Subsequent *alignment* through shared recall or narration can be modeled as a partial matching

$$\mathcal{A}(F_i, F_j) \in [0, 1],$$

representing the degree of shared meaning.

5.4 Dynamics and Proximity (Helix)

Repeated reciprocal intersection events reinforce existing fragments or generate successor fragments. At a macroscopic level, this process increases the *degree of intertwining* between life rays. Operationally, a *twist index* may be updated incrementally after each event:

$$\mathrm{twist}_{ij} \leftarrow \mathrm{twist}_{ij} + \beta_I \bar{I} + \beta_D D + \beta_R \bar{R} + \beta_F,$$

where $\bar{I}$ and $\bar{R}$ aggregate both perspectives (e.g. mean values), and $\beta_\bullet$ denote domain weights.

5.5 Failure and Boundary Cases

- *Sub-threshold events:* Insufficient intensity or novelty $\Rightarrow$ no stable fragment.
- *Interference:* Competing events $\Rightarrow$ distorted or fragmented encoding.
- *Reframing:* Later information alters contextual fit $\Rightarrow$ re-weighting of ℓ_k and s_k.

6 Salience and Fragment Integration

6.1 Salience Function

The imprint strength of an event on an agent's memory is quantified by the *salience* of the resulting fragment. For an agent k, salience is modeled as a weighted sum:

$$s_k = w_I I_k + w_D D + w_R R_k + w_F F_k^* + w_N N_k - w_X X_k, \quad w_\bullet \geq 0.$$

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Here, I_k denotes the perceived interaction intensity, D the duration of the event, R_k the experienced reciprocity, F_k^* the frequency of similar prior events, N_k a novelty factor, and X_k represents interference or competing influences. The weights $w_\bullet$ encode agent-specific sensitivity profiles.

This formulation reflects the assumption that not all encounters contribute equally to memory formation. Salience governs which fragments dominate the fragment chain, thereby shaping memory depth, fragment density, and later resonance effects.

6.2 Integration Operator

While transduction generates individual fragments from shared events, biographical coherence arises through their integration into the existing fragment chain. We model this process by an integration operator acting on the memory structure of a single agent k:

$$A_k : (\mathcal{F}_k(t), F_k) \longrightarrow \mathcal{F}_k(t + \varepsilon).$$

In its minimal form, the operator reduces to sequence concatenation:

$$A_k(\mathcal{F}_k(t), F_k) = \mathcal{F}_k(t) \parallel F_k.$$

The operator A_k modifies the fragment chain by incorporating the new fragment F_k into the existing structure. Depending on contextual and semantic relations, this may involve reinforcement, attenuation, or reweighting of existing fragments.

7 Resonant Asymmetry (Humor)

7.1 Phenomenological Origin

In Part I of the *Life Ray Theory*, humor was introduced at a phenomenological level. Its central mechanisms were described conceptually, without formal mathematical specification.

The core assumptions were the following:

- Humor arises from an asynchrony between an expected meaning time t_e and the actual realization time t_r.
- Prior to semantic resolution, the life ray exhibits a temporary deviation, perceived as a *resonant asymmetry*.

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- The delayed realization produces a local densification of the fragment structure, determining the subjective intensity of the humorous moment.
- The interpreting life ray conceptually *winds* around the punchline's origin before the effective intersection occurs.

This leads to the guiding intuition:

> **Humor is a local increase of energetic density in the fragment configuration, caused by temporally shifted intersection points of two life rays.**

In Part I, this description was intentionally left without formalization or visualization in order to preserve ontological clarity.

7.2 Temporal Asymmetry

The defining structural condition for humor is a temporal divergence between expectation and realization:

$$t_e \neq t_r.$$

The greater this divergence, the more pronounced the pre-resolution fragment accumulation becomes, up to a cognitive or affective saturation threshold.

7.3 Geometric Interpretation

Formally, resonant asymmetry may be interpreted as a spiral-like trajectory of the interpreting ray prior to intersection. The number of windings reflects repeated prediction updates and micro-fragment accumulation. Let W denote the number of effective windings before realization.

7.4 First Humor Metric

As a first approximation, the intensity of humor may be expressed as

$$A_{\text{humor}} \propto \frac{W}{|t_r - t_e|},$$

where W denotes the number of pre-intersection windings and $|t_r - t_e|$ the temporal divergence between expectation and realization.

This expression captures two central intuitions: (i) repeated interpretive cycling (larger W) amplifies the humorous effect, and (ii) increasing temporal delay reduces coherence and thus diminishes humor. In practice, the divergence of the expression for $|t_r - t_e| \to 0$

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is understood to be limited by cognitive saturation effects; consequently, this ratio may be replaced in later models by a normalized or saturating function.

7.5 Position within the Formal Framework

With the introduction of salience and integration, the Life Ray Theory becomes formally tractable: events acquire operative structure, fragments obtain a well-defined internal representation, and interactions receive a mathematical signature.

Within this formal setting, humor constitutes a special dynamical case of resonant interaction. It emerges when high fragment salience coincides with controlled temporal divergence. In later parts, this formalization will support the empirical parameterization of salience and temporal divergence (Part III) and the development of simulations and applied systems (Part IV).

Author's Note and Disclosure

This essay represents an independent conceptual work. All theoretical ideas, models, definitions, and formal structures were conceived, developed, and decided exclusively by the author.

An AI-based writing assistant (ChatGPT by OpenAI) was used solely for the purpose of feedback and critical review, in particular with respect to clarity, structure, consistency of notation, and linguistic formulation. The assistant was further employed for minor language editing, translation, and formatting suggestions.

At no point did the AI system contribute original theoretical content, conceptual design, or mathematical structure. All intellectual responsibility for the content, interpretation, and validity of the presented theory remains entirely with the author.